S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering

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SEMINAR AND PROJECT INITIATION COURSE

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Topic : Phishing Website Detection System

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Abstract

- Phishing means making a fake website that looks like a real one (like a bank or Instagram) to steal passwords and money.
- Many people cannot tell a fake website from a real one.
- Our project checks a website link and tells the user if it is Safe or Phishing.
- It uses Machine Learning: the computer learns from many real and fake websites.

1. Introduction

- **What is phishing?** A trick to steal personal information by pretending to be a trusted website.
- **How does it spread?** Fake links come by SMS, WhatsApp, e-mail and social media.
- **Example:** A message says "Your bank account is blocked, click here". The link opens a fake bank page.
- **Why is it dangerous?** People lose money and their accounts get hacked.
- **Our solution:** A system that checks the link and warns the user.

2. Literature Survey

Sr.	Author and Year	Method used	Finding
1	Zhang, Hong and Cranor, 2007 [1]	CANTINA: checks the words on the web page (TF-IDF) plus some extra clues	Finds fake pages by their content
2	Ma et al., 2009 [2]	Checks the link (URL) and host details using machine learning	Finds bad websites without a blacklist
3	Mohammad et al., 2014 [3]	Neural network that learns from website features (UCI dataset)	Network adjusts to new phishing patterns
4	Sahingoz et al., 2019 [4]	Compared 7 algorithms using words found in the URL	Random Forest was best: 97.98% accuracy
5	Korkmaz et al., 2020 [5]	Compared 8 algorithms on 3 datasets using URL analysis	Machine learning is useful for new fake sites
6	Garera et al., 2007 [6]	Studied clues in phishing links and used logistic regression	Found about 97% of phishing links with few false alarms

2. Literature Survey

Sr.	Author and Year	Method used	Finding
7	Abu-Nimeh et al., 2007 [7]	Compared 6 machine learning methods on phishing e-mails	No single method was best in every test
8	Sheng et al., 2009 [8]	Checked how fast phishing blacklists add new fake sites	Blacklists miss most new fake sites in the first hours
9	Whittaker et al., 2010 [9]	Google system that automatically classifies pages using page and link features	Found most phishing pages with very few false alarms
10	Marchal et al., 2014 [10]	PhishStorm: checks how the words in a link are related to each other	About 95% accuracy using only the link
11	Bahnsen et al., 2017 [11]	Deep learning (LSTM) learns directly from the characters of the URL	Did better than Random Forest with hand-made clues
12	Le et al., 2018 [12]	URLNet: deep learning (CNN) on both characters and words of the URL	Better than older methods at finding bad links

3. Information to seminar topic

How phishing works

1. Attacker copies a real website.
2. Sends the fake link to people.
3. Person enters password or card number.
4. Attacker steals the data.

Common types

- Email phishing – fake emails
- Smishing – fake SMS
- Vishing – fake phone calls

3.1 How we find fake websites

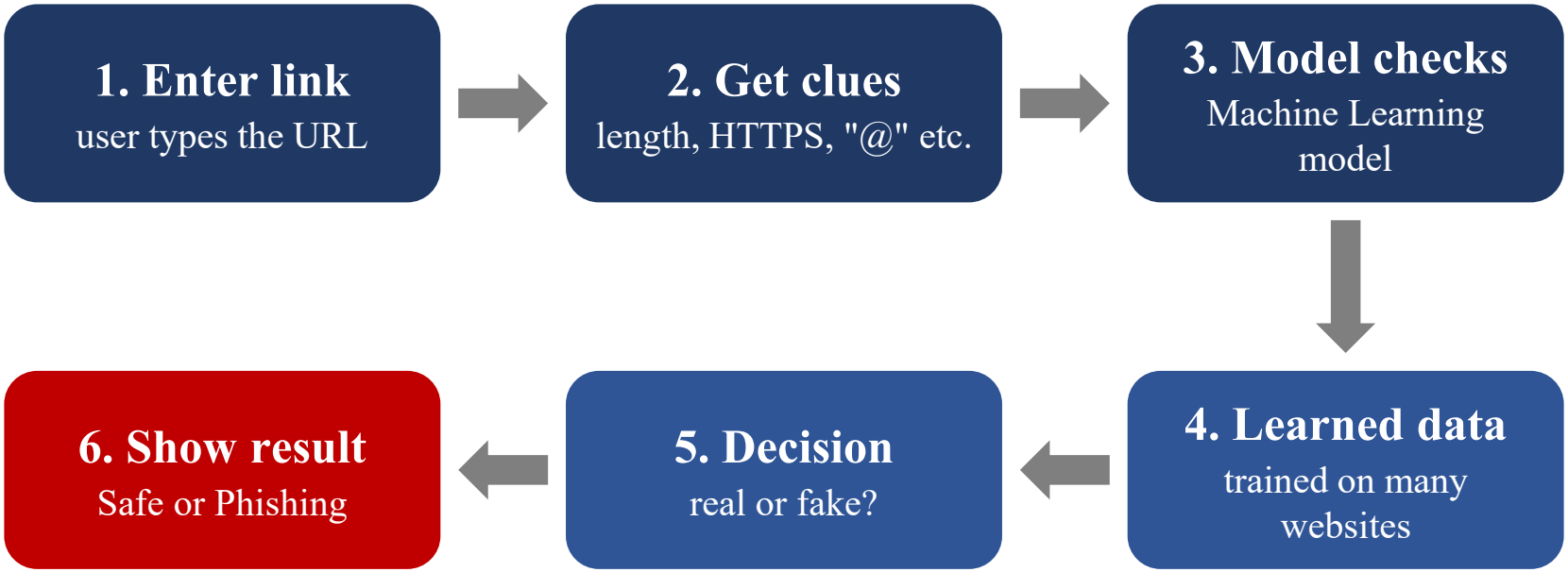
We look at some clues in the link and website:

- **Very long link** – fake links are often long.
- **Has "@" symbol or IP address** – real sites do not use these.
- **No HTTPS (no lock)** – not secure.
- **Too many dots or "-"** – like paypal-login-secure.com
- **New domain** – fake sites are made only a few days ago.

Each clue is turned into a number and given to the computer.

3.2 How our system works

Steps of the system



Training: first we give the computer many real and fake websites so it can learn.

3.3 Machine Learning used

Decision Tree

- Asks yes/no questions, like a flow chart. Example: Is the link very long? Is there "@"?

Random Forest (we use this)

- Many decision trees vote together. The answer with most votes wins, so result is more correct.

Accuracy

- Out of 100 websites, how many are found correctly.

4. Advantages / Disadvantages

Advantages

- Finds new fake websites.
- Gives result in seconds.
- Saves users from fraud.
- Easy to use.
- Can become a browser extension.

Disadvantages

- Needs a lot of data to learn.
- Can make mistakes sometimes.
- Attackers keep changing tricks.
- Needs internet connection.
- Needs updating regularly.

5. Conclusion

- Phishing websites are a big danger for online users.
- Our system checks a link and says Safe or Phishing.
- It uses Machine Learning (Random Forest) to learn from data.
- It can protect people from losing money and data.
- Future: make a Chrome extension and a mobile app.

6. Project Initiation

- **Project title:** Phishing Website Detection System
- **Subject:** Seminar and Project Initiation (SPI – 315003)
- **Group members:** Mehul Pagare, Ishan Pagar, Manish Patil, Sarang Jathar
- **Next slides:** problem, scope, objectives, stakeholders, plan and resources.

6.1 Description of problem statement

Problem

- Fake websites look exactly like real ones.
- Users click on them and lose money or data.
- Old blacklist methods cannot catch new fake sites.

Problem statement

To make a system that checks a website link and tells whether it is Safe or Phishing using Machine Learning.

6.2 Scope and objectives

Objectives

- To understand what phishing is.
- To collect data of real and fake websites.
- To train a Machine Learning model.
- To show result: Safe or Phishing.

Scope

- Checks website links (URLs) given by the user.
- Can be extended to a browser extension later.

6.3 Stake holder

Stakeholder	How they are related
Internet users	Use the system to check links
Students (our team)	Make and present the project
Project guide / teacher	Guides and checks our work
HOD / Department	Gives approval
Banks and shops	Safe from fake website fraud

6.4 Seminar execution plan

Step	Work to do	Time
1	Select topic and study phishing	Week 1
2	Literature survey (read papers)	Week 2
3	Collect website data (dataset)	Week 3
4	Clean and prepare the data	Week 4
5	Find clues (features) from links	Week 5
6	Train the Machine Learning model	Week 6
7	Test the model and check accuracy	Week 7
8	Make a simple web screen	Week 8
9	Final testing and corrections	Week 9
10	Make report and presentation	Week 10

6.5 Platform / Equipment / Resources

Hardware

- Laptop / computer with 4 GB RAM and internet

Software

- Python and Jupyter Notebook (or VS Code)
- Libraries: pandas, scikit-learn

Data (Dataset)

- UCI Phishing Websites dataset (free)
- PhishTank (list of fake websites)

Bibliography

Books

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- Phishing Dark Waters – Christopher Hadnagy and Michele Fincher, Wiley.

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Thank you